

§2.2: andwiched Singularities, Extended Graphs, and Decorated Germs

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1. PROPOSITION II (AS SEEN FROM THE PAPER)

Proposition 1.1. *Let the graph G' be a negative definite plumbing tree, and P' the corresponding plumbing of disk bundles over rational curves. Suppose that the boundary of the plumbing P' is S^3 ; equivalently, G' encodes a configuration of rational curves that can be blown down to a smooth point. For each (-1) vertex, let $\tilde{C}_j$ be a complex disk intersecting the corresponding (-1) sphere in P' transversally once. Let $C_1, \dots, C_m$ be the images of $\tilde{C}_1, \dots, \tilde{C}_m$ under blowing down the configuration G' . Then the following are equivalent:*

1. *Each C_j is smooth.*
2. *There exists exactly one $v'_0 \in G'$ such that $v'_0 \cdot v'_0 + a(v'_0) = -1$, and $v' \cdot v' + a(v') = 0$ for all $v' \neq v'_0$.*

As before, $v' \cdot v'$ denotes the self-intersection of a vertex $v' \in G'$, and $a(v')$ its valence.

2. DEFINITIONS

Definition 2.1. Let G' be a plumbing tree. A **plumbing** is a geometric operation that constructs a 4-dimensional manifold P' by gluing together D^2 -bundles over 2-dimensional spheres (S^2 , or rational curves).

- Each vertex $v' \in G'$ represents a 2-sphere embedded in P' .
- Each edge in G' dictates that the corresponding two spheres intersect transversally exactly once.
- The negative definiteness of the intersection matrix associated with G' ensures that this entire core configuration can be contracted (blown down) to a single point.

Definition 2.2. A (-1) **vertex** in a plumbing tree G' represents an embedded rational curve (sphere) whose self-intersection number is exactly -1 ($v' \cdot v' = -1$).

- Geometrically, this means the sphere can be smoothly blown down (contracted) in the ambient 4-manifold via Castelnuovo's criterion without introducing a singularity.
- In this setup, each (-1) vertex is assigned a small transverse complex disk $\tilde{C}_j$ that punctures the sphere cleanly at a single point.

Definition 2.3. A curve $C_j \subset \mathbb{C}^2$ is **smooth** if it is a embedded complex submanifold of $\mathbb{C}^2$. Geometrically, this means that after the blow-down process collapses the network of spheres into a singular point, the resulting curve image C_j :

- Has a well-defined tangent space at every point.
- Contains no self-intersections (nodes), sharp kinks (cusps), or pinched points.

3. ANALYSIS

The framework operates by transitioning between a 4-dimensional plumbing structure and a flat complex space $\mathbb{C}^2$ via a contraction process:

- **The Ambient Space (P'):** We begin with a 4-dimensional manifold P' built by a plumbing process. This entire 4D space is bounded by, or wrapped inside, a 3-dimensional sphere skin ($\partial P' = S^3$).
- **The Spherical Blueprint (G'):** Enclosed strictly within P' is a complex network of 2-dimensional spheres (S^2 , or rational curves). The plumbing tree G' serves as the precise architectural blueprint, explicitly mapping out how these spheres intersect one another.
- **The Contraction Map (Blow-Down):** Because the configuration graph G' is strictly negative definite, we can mathematically contract or "blow down" this network. This operation crushes the entire extended configuration of 2-spheres down into a singular point, yielding the flat ambient space $\mathbb{C}^2$.

To track how smooth objects behave under this violent collapse, we introduce auxiliary complex structures:

- **Transverse Intersection:** Before collapsing the space, we place small, flat complex disks $\tilde{C}_j$ inside P' . Each disk is positioned to cleanly pierce a specific (-1) sphere—meaning it cuts straight through the sphere at a transversal angle exactly once.
- **Image Construction:** When the blow-down map crushes the underlying network of spheres into a singular point, these transverse disks $\tilde{C}_j$ are pushed forward into $\mathbb{C}^2$. Their resulting image curves are denoted as $C_1, \dots, C_m \subset \mathbb{C}^2$.

The core utility of Proposition 2.2 is that it dictates a perfect equivalence between a global geometric property of these images and a local numerical constraint on the graph topology:

- **Geometric Smoothness Condition:** Every single pushed-forward image curve C_j must be perfectly smooth. This means that after the collapse, the curves retain a well-defined tangent space and develop no sharp kinks (cusps), self-intersections (nodes), or pinched points.
- **Combinatorial Valence Condition:** This absolute smoothness is topologically guaranteed if and only if the plumbing tree G' satisfies a highly restrictive numerical balance:

1. There exists exactly one distinguished, unique "special" vertex $v'_0 \in G'$ where its self-intersection weight plus its valence (the total number of neighboring vertices it connects to, $a(v'_0)$) equals exactly -1 :

$$v'_0 \cdot v'_0 + a(v'_0) = -1$$

2. At every other vertex $v' \neq v'_0$ throughout the rest of the tree, this exact numerical combination must sum to zero:

$$v' \cdot v' + a(v') = 0$$

This exact combinatorial symmetry is not accidental; it directly characterizes the satisfying requirements of a **reduced fundamental cycle**. This condition explicitly guarantees that the resulting configuration corresponds geometrically to a "minimal rational singularity".

4. MATHEMATICAL OUTPUT

Let $C = C_1 \cup \dots \cup C_m$ be a union of smooth complex curves in $\mathbb{C}^2$ intersecting at the origin $(0, 0)$. To separate these curves topologically and build the plumbing tree G' , we perform a sequence of algebraic blow-ups at their intersection points.

1. *Definition of the Blow-Up Coordinate Charts.* The blow-up of $\mathbb{C}^2$ at the origin, denoted $\pi : \tilde{\mathbb{C}}^2 \rightarrow \mathbb{C}^2$, replaces the origin with a projective line $\mathbb{P}^1$ (the exceptional divisor E). This space is covered by two local coordinate charts:

1. **The x -chart:** Coordinates are (x, u) , where the original coordinate y is substituted by:

$$y = u \cdot x \tag{1}$$

In this chart, the exceptional divisor E is given by the locus $\{x = 0\}$. The coordinate u represents the slope of the lines passing through the origin.

2. **The y -chart:** Coordinates are (v, y) , where the original coordinate x is substituted by:

$$x = v \cdot y \quad (2)$$

In this chart, the exceptional divisor E is given by the locus $\{y = 0\}$. The coordinate v represents the inverse slope.

2. *Explicit Example 1: Transverse Intersection (Separated in 1 Step).* Consider two smooth curves intersecting transversally at the origin, defined by the total vanishing locus of $f(x, y) = y^2 - x^2 = 0$. This represents the union of lines $C_1 : y - x = 0$ and $C_2 : y + x = 0$.

Let us apply the x -chart substitution $y = ux$:

$$\begin{aligned} f(x, ux) &= (ux)^2 - x^2 \\ &= u^2 x^2 - x^2 \\ &= x^2(u^2 - 1) \\ &= x^2(u - 1)(u + 1) = 0 \end{aligned}$$

Analysis of the Factored Expression:

- **Exceptional Divisor (E_1):** The factored term $x^2 = 0$ corresponds to the exceptional divisor $x = 0$ created by the blow-up.
- **Proper Transforms ($\tilde{C}_j$):** The remaining part $(u - 1)(u + 1) = 0$ defines the proper transforms of our original curves:

$$\tilde{C}_1 : u = 1, \quad \tilde{C}_2 : u = -1$$

To find where these curves live on the exceptional divisor, we look at their intersection points with E_1 (setting $x = 0$):

- $\tilde{C}_1 \cap E_1 \implies (x, u) = (0, 1)$
- $\tilde{C}_2 \cap E_1 \implies (x, u) = (0, -1)$

Because $1 \neq -1$, these two points are completely distinct on the sphere E_1 . The curves have been successfully **separated** in a single step.

3. *Explicit Example 2: Tangent Intersection (Requiring Repeated Blow-Ups).* Now consider a case where the curves are tangent at the origin, such as a line $C_1 : y = 0$ and a parabola $C_2 : y = x^2$. The total configuration equation is $f(x, y) = y(y - x^2) = 0$.

Blow-Up 1 (x -chart, $y = ux$): Substitute $y = ux$ into the equation:

$$\begin{aligned} f(x, ux) &= (ux)(ux - x^2) \\ &= ux^2(u - x) = 0 \end{aligned}$$

- **Exceptional Divisor (E_1):** Given by $x^2 = 0 \implies x = 0$.
- **Proper Transforms:** Given by the remaining locus $u(u - x) = 0$, where $\tilde{C}_1 : u = 0$ and $\tilde{C}_2 : u = x$.

Let's check their intersection with E_1 by setting $x = 0$:

- $\tilde{C}_1 \cap E_1 \implies (0, 0)$
- $\tilde{C}_2 \cap E_1 \implies (0, 0)$

Because both curves pass through the exact same point $(x, u) = (0, 0)$ on E_1 , the loci are **not yet separated**. As the proof text states, we must blow up again at this new common intersection point.

Blow-Up 2 (New x -chart, $u = w \cdot x$): We now treat (x, u) as our base flat space and blow up its origin $(0, 0)$. We define a new coordinate w such that $u = wx$. Substituting this into the proper transform equation $u(u - x) = 0$:

$$(wx)(wx - x) = wx^2(w - 1) = 0$$

- **New Exceptional Divisor (E_2):** Given by $x^2 = 0 \implies x = 0$.
- **New Proper Transforms:** Given by $w(w - 1) = 0$, meaning $\tilde{C}_1 : w = 0$ and $\tilde{C}_2 : w = 1$.
Let's check their intersection with the newly introduced exceptional divisor E_2 (setting $x = 0$):
- $\tilde{C}_1 \cap E_2 \implies (x, w) = (0, 0)$
- $\tilde{C}_2 \cap E_2 \implies (x, w) = (0, 1)$

Since $0 \neq 1$, the proper transforms are now localized at completely distinct points on E_2 . The loci are fully separated.

Combinatorial Analysis: The Reduced Fundamental Cycle and the Valence Formula

To understand why the combinatorial condition $v' \cdot v' + a(v') = 0$ (or -1) emerges, we connect it to the definition of a **Reduced Fundamental Cycle (RFC)** for a minimal rational singularity.

1. *The Anti-Nef Property of the Fundamental Cycle.* Let $\{E_i\}$ be the set of exceptional divisors (spheres) corresponding to the vertices v_i in the plumbing tree G' . By definition, the fundamental cycle Z is the minimal positive topological cycle satisfying the anti-nef intersection property:

$$Z \cdot E_i \leq 0 \quad \text{for all } i \tag{3}$$

When the fundamental cycle is **reduced**, every exceptional divisor appears with a multiplicity coefficient of exactly 1. Therefore, we can write the cycle explicitly as:

$$Z = \sum_j E_j \tag{4}$$

2. *Deriving the Intersection Formula.* Now, let us calculate the dot product $Z \cdot E_i$ explicitly by expanding the sum:

$$\begin{aligned} Z \cdot E_i &= \left(\sum_j E_j \right) \cdot E_i \\ &= (E_i \cdot E_i) + \sum_{j \neq i} (E_j \cdot E_i) \end{aligned}$$

Let us translate these two algebraic terms directly into the graph properties of our plumbing tree G' :

- **Self-Intersection:** The term $E_i \cdot E_i$ is the self-intersection number of the sphere, represented in the graph as the weight $v_i \cdot v_i$.
- **Mutual Intersection (Valence):** The geometric intersection number $E_j \cdot E_i$ equals 1 if the spheres intersect transversally (meaning vertices v_i and v_j share an edge in G') and 0 otherwise. Therefore, the sum $\sum_{j \neq i} (E_j \cdot E_i)$ counts exactly how many neighbors vertex v_i has, which is its valence $a(v_i)$.

Substituting these graph definitions back into the dot product gives:

$$Z \cdot E_i = v_i \cdot v_i + a(v_i) \tag{5}$$

3. *Matching the Proposition Conditions.* According to Proposition 2.2, the graph must satisfy two specific values for this combination, both of which perfectly satisfy the anti-nef requirement ($Z \cdot E_i \leq 0$):

1. **For the Root Vertex (v'_0):**

$$Z \cdot E_0 = v'_0 \cdot v'_0 + a(v'_0) = -1 \leq 0$$

This means the fundamental cycle strictly pairs negatively with the root divisor.

2. **For all other interior/non-root vertices ($v' \neq v'_0$):**

$$Z \cdot E_i = v' \cdot v' + a(v') = 0 \leq 0$$

This means the cycle pairs completely neutrally with the rest of the network.

4. *Analysis of the Terminal Leaves.* Let us apply this rule to a terminal leaf vertex v_{leaf} at the outer edge of the plumbing tree:

- By definition, a leaf vertex has exactly one neighbor, meaning its valence is:

$$a(v_{\text{leaf}}) = 1$$

- If this leaf is not the root vertex, it must satisfy the zero-balance equation:

$$v_{\text{leaf}} \cdot v_{\text{leaf}} + a(v_{\text{leaf}}) = 0$$

- Substituting $a(v_{\text{leaf}}) = 1$ into the equation yields:

$$v_{\text{leaf}} \cdot v_{\text{leaf}} + 1 = 0 \implies v_{\text{leaf}} \cdot v_{\text{leaf}} = -1$$

This shows exactly why the outer leaves of the tree must be (-1) spheres. This matches Phase 2 of the proof perfectly: because they carry a self-intersection of -1 , they act as the removable "dominoes" that can be smoothly blown down one by one to collapse the tree back to a smooth point.

To see exactly how the linear chains of (-2) spheres form during the resolution process, we track the local geometric transformations without decomposing them into discrete lists. We begin with a single exceptional divisor, which we denote as E_1 . When a collection of curve branches still clusters tightly together at a single point on E_1 , we are forced to perform a subsequent blow-up directly centered at that specific point.

Performing this geometric operation creates a brand-new exceptional sphere, which we call E_2 . Because E_2 is created fresh from a point blow-up, it intrinsically enters the manifold with a baseline self-intersection weight of -1 . At the exact same time, this new sphere is physically attached to the old sphere E_1 at the location of the old intersection point, which creates a new boundary edge in our plumbing graph. This attachment means that E_1 has gained a new neighbor, so its valence increases by exactly 1. Crucially, because we blew up a point resting directly on the body of E_1 , the self-intersection weight of E_1 drops by exactly 1.

We can look at the numerical balance of E_1 before and after this transformation to see how the graph maintains its structure. If E_1 was initially a boundary leaf with a self-intersection of -1 and a valence of 1, the act of blowing up a point on it drops its self-intersection to -2 and pushes its valence up to 2. If we now evaluate our balance equation for this interior vertex, we find that the new self-intersection added to the new valence yields a perfect cancellation:

$$(E_1 \cdot E_1) + a(E_1) = (-2) + 2 = 0$$

This shows that the old sphere E_1 has been transformed into a stable interior (-2) sphere that satisfies our zero-balance condition. Meanwhile, the new sphere E_2 now inherits the role of the terminal leaf, carrying the active self-intersection weight of -1 and a valence of 1.

If the curves are tangent or require further separation along this branch, we repeat the exact same procedure by blowing up a point on our newest leaf E_2 . This creates a third exceptional sphere, E_3 , with an initial weight of -1 . Just like before, blowing up a point on E_2 drops its weight from -1 down to -2 , while adding E_3 as a neighbor increases the valence of E_2 from 1 up to 2. The balance equation for E_2 smoothly transitions to zero:

$$(E_2 \cdot E_2) + a(E_2) = (-2) + 2 = 0$$

This iterative process generates a growing linear chain of spheres where every single interior vertex is mathematically locked into a self-intersection of -2 and a valence of 2. The process behaves like an expanding telescope, continually pushing a single active (-1) sphere out at the very tip of the branch while leaving a trail of balanced (-2) spheres behind it. The algorithm terminates only when the blow-up separates the curves completely, leaving them at distinct points $p_1^1, p_2^1, \dots$ along the final (-1) sphere boundary.

Stability of the Tree Structure and the Smoothness Criterion

To understand why the algorithm is guaranteed to terminate and preserve our combinatorial balance equations, we must examine the local geometry of the curve branches using the Jacobian criterion. A curve branch C_j defined by the vanishing of a local holomorphic function $f_j(x, y) = 0$ is smooth at a point if and only if the gradient vector of partial derivatives does not vanish at that point. This non-vanishing gradient means that the curve can be locally parametrized as a regular embedded disk, which forces its local algebraic multiplicity to be exactly one. Because every curve branch possesses a multiplicity of one, any intersection it forms with an exceptional divisor after a blow-up is structurally forced to have a geometric multiplicity of at most one. This bounds the local intersection behavior and guarantees that a single blow-up at an intersection point p_i^1 is completely sufficient to make the proper transform curve branch $\tilde{C}_j$ entirely disjoint from the exceptional divisor B_1 .

Once the collection of curve branches becomes disjoint from a specific exceptional divisor like B_1 , the algorithm ensures that we will never perform another blow-up at any point lying on that divisor again. This creates a critical stability point in the proof because the self-intersection weight and the valence of that vertex are now permanently locked and can be computed explicitly. Suppose the divisor B_1 initially hosts r_1 distinct intersection points where curves or other branches diverge. Blowing up at each of these r_1 points drops the self-intersection weight of B_1 by exactly r_1 units from its baseline. If B_1 is an interior branch of the tree, its initial baseline self-intersection before these separating blow-ups was -1 . Dropping its weight by r_1 means its final, frozen self-intersection weight becomes exactly $-r_1 - 1$.

We can check if this stable state satisfies our invariant conditions by evaluating the valence of B_1 in the final graph topology. Because B_1 is an interior divisor, it is connected to its original parent sphere, and it has now birthed r_1 new child spheres from the blow-ups at its intersection points. This gives it a final neighbor count, or valence, of exactly $r_1 + 1$. Adding this final valence to its frozen self-intersection weight gives:

$$(B_1 \cdot B_1) + a(B_1) = (-r_1 - 1) + (r_1 + 1) = 0$$

This confirms that the interior vertex stabilizes perfectly to satisfy our zero-balance condition. If B_1 happened to be the absolute root of the entire tree, it would have no parent connection, meaning its final valence would simply be r_1 . Combining this with its baseline root weight of -1 minus the r_1 blow-ups yields a final weight of $-r_1 - 1$. Evaluating the balance equation for this root vertex gives:

$$(B_1 \cdot B_1) + a(B_1) = (-r_1 - 1) + r_1 = -1$$

This shows that whether a divisor acts as the root or an interior branch, the smooth multiplicity of the curves guarantees that the graph topology freezes into a state that perfectly satisfies Condition 2 of the proposition. The entire global plumbing tree G' is built by repeating this stabilization process iteratively at every remaining cluster point until all curves are fully separated into disjoint spheres.

The Converse Direction

To complete the analysis of the proposition, we must evaluate the converse argument: proving that if a plumbing tree G' satisfies Condition 2, blowing down the configuration must yield perfectly smooth curves $C_j \subset \mathbb{C}^2$. The structural entry point for this reverse induction relies entirely on the outer topology of the tree. If the tree satisfies the zero-balance equation everywhere except at the root, the only vertices that can carry a self-intersection weight of exactly -1 are the terminal leaves. This is because a leaf vertex by definition possesses a valence of exactly one, meaning its balance equation forces its self-intersection to be $v \cdot v + 1 = 0$, which yields $v \cdot v = -1$.

Because these terminal leaves are exceptional spheres of weight -1 , Castelnuovo's criterion guarantees that they can be smoothly collapsed or blown down to a regular point without introducing any local geometric singularities. We can track the behavior of the transverse disks $\tilde{C}_j$ under this collapse to see why smoothness is preserved. Each disk $\tilde{C}_j$ intersects its corresponding terminal leaf sphere transversally with a geometric multiplicity of exactly one. When the blow-down map shrinks the -1 sphere to a single point, this multiplicity-one intersection ensures that the disk simply slides smoothly onto the adjacent neighboring sphere in the tree. Because the intersection is clean and transverse, the collapsing process does not create any local topological folding, sharp cusps, or self-intersections on the disk transform.

We can track the numerical adjustments of the remaining graph during this collapse to see how the induction propagates inward toward the root. When a terminal leaf is blown down and erased from the graph, its immediate neighbor loses a connection, which causes that neighbor's valence to decrease by exactly one. Simultaneously, the algebraic mechanics of a surface blow-down dictate that collapsing a -1 sphere increases the self-intersection weight of its adjacent neighbor by exactly one. Because the valence drops by one and the self-intersection increases by one, the combined balance equation for that neighboring vertex remains perfectly invariant:

$$\Delta(v \cdot v) + \Delta(a(v)) = (+1) + (-1) = 0$$

This invariance is the core engine of the converse proof. It guarantees that after a leaf is erased, the adjacent interior vertex—which previously had a self-intersection of -2 and a valence of two—gains a weight of -1 and drops to a valence of one. This means the adjacent vertex has now seamlessly inherited the status of a terminal -1 leaf. We can therefore apply this process inductively, knocking away the outer layers of the plumbing tree one by one like dominoes. Because every single collapsing step along the way involves a multiplicity-one intersection on a -1 sphere, smoothness is rigorously preserved at every stage of the reduction. When the entire tree is finally collapsed down to the absolute root and pushed into $\mathbb{C}^2$, the resulting proper transform curves C_j are guaranteed to emerge as perfectly smooth embedded submanifolds.

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